SEMESTER III

Name of Department: - Computer Science and Engineering

1. Subject Code: **TCS 392** Course Title: **Introduction to Cryptography**

2. Contact Hours: **L:** 3 **T**: **P**:

3. Examination Duration (Hrs): **Theory** **Practical**

4. Relative Weight: **CIE** 25 **MSE** 25 **SEE** 50

5. Credits: 3

6. Semester: III

7. Category of Course: **DE**

8. Pre-requisite: NA

9. **Course Outcome**:**	After completion of the course the students will be able to: CO1:Classify security vulnerabilities involved in data communication over Internet and makeuse of classical algorithms to address the vulnerabilities. CO2: Apply symmetric block ciphers to secure data transmission and storage CO3: Analyze the various public key cryptographic systems and usage of hashing CO4 Appreciate the design of Public Key algorithms, mathematical background and make useof the same for data communication and message authentication CO5: Categorize types of viruses, worms, intrusion and decide measures to counter thethreats. CO6: Understand the legal aspects related to Cybercrime, Intellectual Property, Privacy,Ethical Issues.

** Describe the specific knowledge, skills or competencies the students are expected to

10. Details of the Course:

Sl. No.	Contents	Contact Hours
1	**Unit – 1:** Introduction: Computer Security Concepts: The OSI SecurityArchitecture, Security Attacks, Security Services, Security Mechanisms, a Model for Network Security, Standards Cryptography fundamentals and terminology; Cryptanalysis and Brute-Force Attack, Fundamental techniques of cryptography Substitution and Transposition; Classical Ciphers; Basics of Steganography.	8
2	**Unit – 2:** Modern Cryptography: Symmetric Encryption and MessageConfidentiality: Symmetric Encryption Principles, Fiestal structure. Symmetric Block Encryption Algorithms, Simple DES, double DES, Stream Ciphers and RC4, Random and Pseudorandom Numbers.	9
3	**Unit – 3:** Symmetric key distribution using symmetric encryption: A Key Distribution Scenario, Session Key Lifetime, A Transparent Key Control Scheme, Decentralized Key Control, Controlling Key UsageMathematical Background for cryptography: prime number, Euclidean algorithm for GCD, Extended Euclidean algorithm for multiplicative inverse, Euler's totient function, their programming implementation.	10
4	**Unit 4:** Public-Key Cryptography: Public-Key Encryption Structure, Applications for Public-Key Cryptosystems, Requirements for Public-Key Cryptography, The RSA Public-Key Encryption Algorithm. Message Authentication: Approaches to Message Authentication, Authentication Using Conventional Encryption, Message Authentication without Message Encryption, MD5 Hash Algorithm.	9
5	**Unit 5:** System Security: Intruders, Intrusion Detection, Password Management, Types of Malicious Software, Viruses, Virus Countermeasures, Worms and Principles of Firewalls Legal and Ethical Aspects: Cybercrime and Computer Crime, Intellectual Property, Privacy, Ethical Issues.	8
	Total	**44**